SQL- ASSIGNMENT-1 (ITEMS AND PRODCUTS DATABASE)

Basics Questions:-

1. Create two Databases Name :- Brands and Products

- create database Brands;
- create database Products;

2. Create two tables in SQL Server name as ITEMS_TABLE in Brands database and PRODUCT_TABLE in Products database.

- use Brands;
- CREATE TABLE ITEMS_TABLE (
  item_Id INT PRIMARY KEY,
  item_description VARCHAR(100),
  vendor_nos INT,
  vendor_name VARCHAR(100),
  bottle_size INT,
  bottle_price DECIMAL(10,2)

  );

- use Products;
- CREATE TABLE PRODUCT_SALES_TABLE (
  Product_Id INT PRIMARY KEY,
  Country VARCHAR(50),
  Product VARCHAR(100),
  Units_Sold DECIMAL(10,2),
  Manufacturing_Price DECIMAL(10,2),
  Sale_Price DECIMAL(10,2),
  Gross_Sales DECIMAL(12,2),
  Sales DECIMAL(12,2),
  COGS DECIMAL(12,2),
  Profit DECIMAL(12,2),
  Date DATE,
  Month_Number INT,
  Month_Name VARCHAR(20),
  Year INT

  );

3. After Creating both the tables Add records in that tables (records are available in ITEMS_TABLE Sheet and PRODUCTS_TABLE Sheet)

- INSERT INTO ITEMS_TABLE (item_Id, item_description, vendor_nos, vendor_name, bottle_size, bottle_price)
  VALUES
  (1, 'Travis Hasse Apple Pie', 305, 'Mhw Ltd', 750, 9.77),
  (2, 'D'aristi Xtabentun', 391, 'Anchor Distilling (preiss Imports)', 750, 14.12),
  (3, 'Hiram Walker Peach Brandy', 370, 'Pernod Ricard Usa/austin Nichols', 1000, 6.50),
  (4, 'Oak Cross Whisky', 305, 'Mhw Ltd', 750, 25.33),
  (5, 'Uv Red(cherry) Vodka', 380, 'Phillips Beverage Company', 200, 1.97),
  (6, 'Heaven Hill Old Style White Label', 259, 'Heaven Hill Distilleries Inc.', 750, 6.37),
  (7, 'Hyde Herbal Liqueur', 194, 'Fire Tail Brands Llc', 750, 5.06),
  (8, 'Dupont Calvados Fine Reserve', 403, 'Robert Kacher Selections', 750, 23.61);
- select * from ITEMS_TABLE;
- Output:

	item_Id	item_description	vendor_nos	vendor_name	bottle_size	bottle_price
1	1	Travis Hasse Apple Pie	305	Mhw Ltd	750	9.77
2	2	D'aristi Xtabentun	391	Anchor Distilling (preiss Imports)	750	14.12
3	3	Hiram Walker Peach Brandy	370	Pernod Ricard Usa/austin Nichols	1000	6.50
4	4	Oak Cross Whisky	305	Mhw Ltd	750	25.33
5	5	Uv Red(cherry) Vodka	380	Phillips Beverage Company	200	1.97
6	6	Heaven Hill Old Style White Label	259	Heaven Hill Distilleries Inc.	750	6.37
7	7	Hyde Herbal Liqueur	194	Fire Tail Brands Llc	750	5.06
8	8	Dupont Calvados Fine Reserve	403	Robert Kacher Selections	750	23.61

- INSERT INTO PRODUCT_SALES_TABLE
  (Product_Id, Country, Product, Units_Sold, Manufacturing_Price, Sale_Price, Gross_Sales, Sales, COGS, Profit, Date, Month_Number, Month_Name, Year)
  VALUES
  (1, 'Canada', 'Carretera', 1618.5, 3, 20, 32370, 32370, 16185, 16185, '2014-01-01', 1, 'January', 2014),
  (2, 'Germany', 'Carretera', 1321, 3, 20, 26420, 26420, 13120, 13300, '2014-01-01', 1, 'January', 2015),
  (3, 'France', 'Carretera', 2178, 3, 15, 32670, 32670, 21970, 10700, '2014-06-01', 6, 'June', 2016),
  (4, 'Germany', 'Carretera', 888, 15, 15, 13320, 13320, 8880, 4440, '2014-06-01', 6, 'June', 2017),
  (5, 'Mexico', 'Carretera', 2470, 3, 15, 37050, 24700, 12350, 24700, '2014-10-01', 10, 'October', 2018),

(6, 'Germany', 'Carretera', 1513, 350, 529.55, 529550, 529550, 185130, 344420, '2014-12-01', 12, 'December', 2019),
(7, 'Germany', 'Montana', 921, 5, 15, 13815, 13815, 9210, 4605, '2020-03-01', 3, 'March', 2020),
(8, 'Canada', 'Montana', 2518, 5, 12, 30216, 30216, 7554, 22662, '2021-06-01', 6, 'June', 2021);

- select * from PRODUCT_SALES_TABLE;
- Output:

	Product_Id	Country	Product	Units_Sold	Manufacturing_Price	Sale_Price	Gross_Sales	Sales	COGS	Profit	Date	Month_Number	Month_Name	Year
1	1	Canada	Carretera	1618.50	3.00	20.00	32370.00	32370.00	16185.00	16185.00	2014-01-01	1	January	2014
2	2	Germany	Carretera	1321.00	3.00	20.00	26420.00	26420.00	13120.00	13300.00	2014-01-01	1	January	2015
3	3	France	Carretera	2178.00	3.00	15.00	32670.00	32670.00	21970.00	10700.00	2014-06-01	6	June	2016
4	4	Germany	Carretera	888.00	15.00	15.00	13320.00	13320.00	8880.00	4440.00	2014-06-01	6	June	2017
5	5	Mexico	Carretera	2470.00	3.00	15.00	37050.00	37050.00	24700.00	12350.00	2024-01-06	6	June	2018
6	6	Germany	Carretera	1513.00	350.00	529.55	529550.00	529550.00	185130.00	344420.00	2014-12-01	12	December	2019
7	7	Germany	Montana	921.00	5.00	15.00	13815.00	13815.00	9210.00	4605.00	2020-03-01	3	March	2020
8	8	Canada	Montana	2518.00	5.00	12.00	30216.00	30216.00	7554.00	22662.00	2021-06-01	6	June	2021

4. Delete those product having the Units Sold 1618.5 , 888 and 2470.

- delete from PRODUCT_SALES_TABLE where Units_Sold IN (1618.5, 888, 2470);
- Output:
  (3 rows affected)
  Completion time: 2026-03-07T14:28:36.6987806+05:30

5. Select all records from the ITEMS_TABLE table.
- select * from ITEMS_TABLE;
- Output:

	item_Id	item_description	vendor_nos	vendor_name	bottle_size	bottle_price
1	1	Travis Hasse Apple Pie	305	Mhw Ltd	750	9.77
2	2	D'aristi Xtabentun	391	Anchor Distilling (preiss Imports)	750	14.12
3	3	Hiram Walker Peach Brandy	370	Pernod Ricard Usa/austin Nichols	1000	6.50
4	4	Oak Cross Whisky	305	Mhw Ltd	750	25.33
5	5	Uv Red(cherry) Vodka	380	Phillips Beverage Company	200	1.97
6	6	Heaven Hill Old Style White Label	259	Heaven Hill Distilleries Inc.	750	6.37
7	7	Hyde Herbal Liqueur	194	Fire Tail Brands Llc	750	5.06
8	8	Dupont Calvados Fine Reserve	403	Robert Kacher Selections	750	23.61

6. Select the item_description and bottle_price for all items in the ITEMS_TABLE table.
- select item_description,bottle_price from ITEMS_TABLE;
- Output:

	item_description	bottle_price
1	Travis Hasse Apple Pie	9.77
2	D'aristi Xtabentun	14.12
3	Hiram Walker Peach Brandy	6.50
4	Oak Cross Whisky	25.33
5	Uv Red(cherry) Vodka	1.97
6	Heaven Hill Old Style White Label	6.37
7	Hyde Herbal Liqueur	5.06
8	Dupont Calvados Fine Reserve	23.61

7. Find the item_description and bottle_price of items where bottle_price is greater than 20.

- select item_description,bottle_price from ITEMS_TABLE where bottle_price > 20;
- Output:

	item_description	bottle_price
1	Oak Cross Whisky	25.33
2	Dupont Calvados Fine Reserve	23.61

8. Select Unique Country from the product_sales table

- select distinct Country from PRODUCT_SALES_TABLE;
- Output:

	Country
1	Canada
2	France
3	Germany
4	Mexico

9. Count the number of Countries in the product_sales table

- select count (DISTINCT Country) as Total_Countries from PRODUCT_SALES_TABLE;
- Output:

	Total_Countries
1	4

10. How Many Countries are there which contain the sales price between 10 to 20

- select distinct Country from PRODUCT_SALES_TABLE where Sale_Price between 10 and 20;
- Output:

	Country
1	Canada
2	France
3	Germany
4	Mexico

Intermediate Questions:-

PRODUCTS_SALES_TABLE:-

1. Find the Total Sale Price and Gross Sales
   - select SUM(Sale_Price) as Total_Sale_Price, SUM(Gross_Sales) AS Total_Gross_Sales from PRODUCT_SALES_TABLE;
   - Output:

	Total_Sale_Price	Total_Gross_Sales
1	641.55	715411.00

2. In which year we have got the highest sales
   - SELECT TOP 1 Year, SUM(Sales) AS Total_Sales FROM PRODUCT_SALES_TABLE
     GROUP BY Year ORDER BY Total_Sales DESC;
   - Output:

	Year	Total_Sales
1	2019	529550.00

3. Which Product having the sales of $ 37,050.00
   - SELECT Product
     FROM PRODUCT_SALES_TABLE
     WHERE Sales = 37050;
   - Output:

	Product
1	Carretera

4. Which Countries lies between profit of $ 4,605 to $ 22 , 662.00

- select Distinct Country from PRODUCT_SALES_TABLE where Profit between 4605 AND 22662;
- Output:

	Country
1	Canada
2	France
3	Germany
4	Mexico

5. Which Product Id having the sales of $ 24 , 700.00
   - SELECT Product_Id
     FROM PRODUCT_SALES_TABLE
     WHERE Sales = 24700;
   - Output:

Product_Id

6. Find the total Units Sold for each Country.
   - SELECT Country, SUM(Units_Sold) AS Total_Units_Sold
     FROM PRODUCT_SALES_TABLE
     GROUP BY Country;
   - Output:

	Country	Total_Units_Sold
1	Canada	4136.50
2	France	2178.00
3	Germany	4643.00
4	Mexico	2470.00

7. Find the average sales for each country
   - SELECT Country, AVG(Sales) AS Average_Sales
     FROM PRODUCT_SALES_TABLE
     GROUP BY Country;
   - Output:

	Country	Average_Sales
1	Canada	31293.000000
2	France	32670.000000
3	Germany	145776.250000
4	Mexico	37050.000000

8. Retrieve all products sold in 2014

- SELECT *
  FROM PRODUCT_SALES_TABLE
  WHERE Year = 2014;
- Output:

	Product_Id	Country	Product	Units_Sold	Manufacturing_Price	Sale_Price	Gross_Sales	Sales	COGS	Profit	Date	Month_Number	Month_Name	Year
1	1	Canada	Carretera	1618.50	3.00	20.00	32370.00	32370.00	16185.00	16185.00	2014-01-01	1	January	2014

9. Find the maximum Profit in the product_sales table.

- SELECT MAX(Profit) AS Maximum_Profit FROM PRODUCT_SALES_TABLE;
- Output:

	Maximum_Profit
1	344420.00

10. Retrieve the records from product_sales where Profit is greater than the average Profit of all records.

- SELECT *
  FROM PRODUCT_SALES_TABLE
  WHERE Profit > (SELECT AVG(Profit) FROM PRODUCT_SALES_TABLE);
- Output:

	Product_Id	Country	Product	Units_Sold	Manufacturing_Price	Sale_Price	Gross_Sales	Sales	COGS	Profit	Date	Month_Number	Month_Name	Year
1	6	Germany	Carretera	1513.00	350.00	529.55	529550.00	529550.00	185130.00	344420.00	2014-12-01	12	December	2019

11. Find the item_description having the bottle size of 750

- SELECT Item_Description FROM ITEMS_TABLE WHERE Bottle_Size = 750;
- Output:

	Item_Description
1	Travis Hasse Apple Pie
2	D'aristi Xtabentun
3	Oak Cross Whisky
4	Heaven Hill Old Style White Label
5	Hyde Herbal Liqueur
6	Dupont Calvados Fine Reserve

12. Find the vendor Name having the vendor_nos 305 , 380 , 391

- SELECT Vendor_Name FROM ITEMS_TABLE WHERE Vendor_Nos IN (305, 380, 391);
- Output:

	Vendor_Name
1	Mhw Ltd
2	Anchor Distilling (preiss Imports)
3	Mhw Ltd
4	Phillips Beverage Company

13. What is total Bottle_price

- SELECT SUM(Bottle_Price) AS Total_Bottle_Price FROM ITEMS_TABLE;
- Output:

	Total_Bottle_Price
1	92.73

14. Make Primary Key to Item_id

- ALTER TABLE ITEMS_TABLE ADD CONSTRAINT PK_ITEM PRIMARY KEY (Item_Id);

15. Which item id having the bottle_price of $ 5.06

- SELECT Item_Id FROM ITEMS_TABLE WHERE Bottle_Price = 5.06;
- Output:

	Item_Id
1	7

Advance Questions:-

1. Apply INNER , FULL OUTER , LEFT JOIN types on both the table

1) INNER JOIN:

- SELECT * FROM ITEMS_TABLE I INNER JOIN PRODUCT_SALES_TABLE P ON I.Item_Id = P.Product_Id;
- Output:

	item_Id	item_description	vendor_nos	vendor_name	bottle_size	bottle_price	Product_Id	Country	Product	Units_Sold	Manufacturing_Price	Sale_Price	Gross_Sales	Sales	COGS
1	2	D'aristi Xtabentun	391	Anchor Distilling (preiss Imports)	750	14.12	2	Germany	Carretera	1321.00	3.00	20.00	26420.00	26420.00	13120.00
2	3	Hiram Walker Peach Brandy	370	Pernod Ricard Usa/austin Nichols	1000	6.50	3	France	Carretera	2178.00	3.00	15.00	32670.00	32670.00	21970.00
3	6	Heaven Hill Old Style White Label	259	Heaven Hill Distilleries Inc.	750	6.37	6	Germany	Carretera	1513.00	350.00	529.55	529550.00	529550.00	185130.00
4	7	Hyde Herbal Liqueur	194	Fire Tail Brands Llc	750	5.06	7	Germany	Montana	921.00	5.00	15.00	13815.00	13815.00	9210.00
5	8	Dupont Calvados Fine Reserve	403	Robert Kacher Selections	750	23.61	8	Canada	Montana	2518.00	5.00	12.00	30216.00	30216.00	7554.00

Profit	Date	Month_Number	Month_Name	Year
13300.00	2014-01-01	1	January	2015
10700.00	2014-06-01	6	June	2016
344420.00	2014-12-01	12	December	2019
4605.00	2020-03-01	3	March	2020
22662.00	2021-06-01	6	June	2021

2) LEFT JOIN :

- SELECT *
  FROM ITEMS_TABLE I
  LEFT JOIN PRODUCT_SALES_TABLE P
  ON I.Item_Id = P.Product_Id;
- Output:

	item_Id	item_description	vendor_nos	vendor_name	bottle_size	bottle_price	Product_Id	Country	Product	Units_Sold	Manufacturing_Price	Sale_Price	Gross_Sales	Sales	COGS
1	1	Travis Hasse Apple Pie	305	Mhw Ltd	750	9.77	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
2	2	D'aristi Xtabentun	391	Anchor Distilling (preiss Imports)	750	14.12	2	Germany	Carretera	1321.00	3.00	20.00	26420.00	26420.00	13120.00
3	3	Hiram Walker Peach Brandy	370	Pernod Ricard Usa/austin Nichols	1000	6.50	3	France	Carretera	2178.00	3.00	15.00	32670.00	32670.00	21970.00
4	4	Oak Cross Whisky	305	Mhw Ltd	750	25.33	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
5	5	Uv Red(cherry) Vodka	380	Phillips Beverage Company	200	1.97	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
6	6	Heaven Hill Old Style White Label	259	Heaven Hill Distilleries Inc.	750	6.37	6	Germany	Carretera	1513.00	350.00	529.55	529550.00	529550.00	185130.00
7	7	Hyde Herbal Liqueur	194	Fire Tail Brands Llc	750	5.06	7	Germany	Montana	921.00	5.00	15.00	13815.00	13815.00	9210.00
8	8	Dupont Calvados Fine Reserve	403	Robert Kacher Selections	750	23.61	8	Canada	Montana	2518.00	5.00	12.00	30216.00	30216.00	7554.00

Profit	Date	Month_Number	Month_Name	Year
NULL	NULL	NULL	NULL	NULL
13300.00	2014-01-01	1	January	2015
10700.00	2014-06-01	6	June	2016
NULL	NULL	NULL	NULL	NULL
NULL	NULL	NULL	NULL	NULL
344420.00	2014-12-01	12	December	2019
4605.00	2020-03-01	3	March	2020
22662.00	2021-06-01	6	June	2021

3) OUTER JOIN :

- SELECT *
  FROM ITEMS_TABLE I
  FULL OUTER JOIN PRODUCT_SALES_TABLE P
  ON I.Vendor_Nos = P.Product_Id;
- Output:

	item_Id	item_description	vendor_nos	vendor_name	bottle_size	bottle_price	Product_Id	Country	Product	Units_Sold	Manufacturing_Price	Sale_Price	Gross_Sales	Sales	COGS
1	1	Travis Hasse Apple Pie	305	Mhw Ltd	750	9.77	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
2	2	D'aristi Xtabentun	391	Anchor Distilling (preiss Imports)	750	14.12	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
3	3	Hiram Walker Peach Brandy	370	Pernod Ricard Usa/austin Nichols	1000	6.50	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
4	4	Oak Cross Whisky	305	Mhw Ltd	750	25.33	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
5	5	Uv Red(cherry) Vodka	380	Phillips Beverage Company	200	1.97	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
6	6	Heaven Hill Old Style White Label	259	Heaven Hill Distilleries Inc.	750	6.37	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
7	7	Hyde Herbal Liqueur	194	Fire Tail Brands Llc	750	5.06	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
8	8	Dupont Calvados Fine Reserve	403	Robert Kacher Selections	750	23.61	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
9	NULL	NULL	NULL	NULL	NULL	NULL	2	Germany	Carretera	1321.00	3.00	20.00	26420.00	26420.00	13120.00
10	NULL	NULL	NULL	NULL	NULL	NULL	3	France	Carretera	2178.00	3.00	15.00	32670.00	32670.00	21970.00
11	NULL	NULL	NULL	NULL	NULL	NULL	6	Germany	Carretera	1513.00	350.00	529.55	529550.00	529550.00	185130.00
12	NULL	NULL	NULL	NULL	NULL	NULL	7	Germany	Montana	921.00	5.00	15.00	13815.00	13815.00	9210.00
13	NULL	NULL	NULL	NULL	NULL	NULL	8	Canada	Montana	2518.00	5.00	12.00	30216.00	30216.00	7554.00

Query executed successfully. DESKTOP-JL1DVTD\SQLEXPRESS ... DESKTOP-JL1DVTD\Velcy ... Brands 00:00:00 13

Profit	Date	Month_Number	Month_Name	Year
NULL	NULL	NULL	NULL	NULL
NULL	NULL	NULL	NULL	NULL
NULL	NULL	NULL	NULL	NULL
NULL	NULL	NULL	NULL	NULL
NULL	NULL	NULL	NULL	NULL
NULL	NULL	NULL	NULL	NULL
NULL	NULL	NULL	NULL	NULL
NULL	NULL	NULL	NULL	NULL
13300.00	2014-01-01	1	January	2015
10700.00	2014-06-01	6	June	2016
344420.00	2014-12-01	12	December	2019
4605.00	2020-03-01	3	March	2020
22662.00	2021-06-01	6	June	2021

2. Find the item_description and Product having the gross sales of 13,320.00

- SELECT I.Item_Description, P.Product
  FROM PRODUCT_SALES_TABLE P
  JOIN ITEMS_TABLE I
  ON I.Item_Id = P.Product_Id
  WHERE P.Gross_Sales = 13320;
- Output:

	Item_Description	Product

3. Split the Item_description Column into Columns Item_desc1 and Item_desc2

- SELECT
  Item_Description,
  LEFT(Item_Description, CHARINDEX(' ', Item_Description) - 1) AS Item_desc1,
  SUBSTRING(Item_Description, CHARINDEX(' ', Item_Description) + 1,
  LEN(Item_Description)) AS Item_desc2
  FROM ITEMS_TABLE;
- Output:

	Item_Description	Item_desc1	Item_desc2
1	Travis Hasse Apple Pie	Travis	Hasse Apple Pie
2	D'aristi Xtabentun	D'aristi	Xtabentun
3	Hiram Walker Peach Brandy	Hiram	Walker Peach Brandy
4	Oak Cross Whisky	Oak	Cross Whisky
5	Uv Red(cherry) Vodka	Uv	Red(cherry) Vodka
6	Heaven Hill Old Style White Label	Heaven	Hill Old Style White Label
7	Hyde Herbal Liqueur	Hyde	Herbal Liqueur
8	Dupont Calvados Fine Reserve	Dupont	Calvados Fine Reserve

4. Find the top 3 most expensive items in the ITEMS_TABLE table.

- SELECT TOP 3 * FROM ITEMS_TABLE ORDER BY Bottle_Price DESC;
- Output:

	item_Id	item_description	vendor_nos	vendor_name	bottle_size	bottle_price
1	4	Oak Cross Whisky	305	Mhw Ltd	750	25.33
2	8	Dupont Calvados Fine Reserve	403	Robert Kacher Selections	750	23.61
3	2	D'aristi Xtabentun	391	Anchor Distilling (preiss Imports)	750	14.12

5. Find the total Gross Sales and Profit for each Product in each Country.

- SELECT
  Country,
  Product,
  SUM(Gross_Sales) AS Total_Gross_Sales,
  SUM(Profit) AS Total_Profit
  FROM PRODUCT_SALES_TABLE
  GROUP BY Country, Product;
- Output:

	Country	Product	Total_Gross_Sales	Total_Profit
1	France	Carretera	32670.00	10700.00
2	Germany	Carretera	555970.00	357720.00
3	Canada	Montana	30216.00	22662.00
4	Germany	Montana	13815.00	4605.00

6. Find the vendor_name and item_description of items with a bottle_size of 750 and bottle_price less than 10.

- SELECT Vendor_Name, Item_Description
  FROM ITEMS_TABLE
  WHERE Bottle_Size = 750
  AND Bottle_Price < 10;
- Output:

	Vendor_Name	Item_Description
1	Mhw Ltd	Travis Hasse Apple Pie
2	Heaven Hill Distilleries Inc.	Heaven Hill Old Style White Label
3	Fire Tail Brands Llc	Hyde Herbal Liqueur

7. Find the Product with the highest Profit in 2019.

- SELECT TOP 1 Product, Profit FROM PRODUCT_SALES_TABLE WHERE Year = 2019
  ORDER BY Profit DESC;

- Output:

	Product	Profit
1	Carretera	344420.00

8. Retrieve the Product_Id and Country of all records where the Profit is at least twice the COGS.

- SELECT Product_Id, Country
  FROM PRODUCT_SALES_TABLE
  WHERE Profit >= (2 * COGS);
- Output:

	Product_Id	Country
1	8	Canada

9. Find the Country that had the highest total Gross Sales in 2018

- SELECT TOP 1 Country, SUM(Gross_Sales) AS Total_Gross
  FROM PRODUCT_SALES_TABLE
  WHERE Year = 2018
  GROUP BY Country
  ORDER BY Total_Gross DESC;
- Output:

	Country	Total_Gross

10. Calculate the total Sales for each Month Name across all years.

- SELECT Month_Name, SUM(Sales) AS Total_Sales
  FROM PRODUCT_SALES_TABLE
  GROUP BY Month_Name;
- Output:

	Month_Name	Total_Sales
1	December	529550.00
2	January	26420.00
3	June	62886.00
4	March	13815.00

11. List the item_description and vendor_name for items whose vendor_nos exists more than once in the ITEMS_TABLE table.

- SELECT Item_Description, Vendor_Name FROM ITEMS_TABLE WHERE Vendor_Nos IN (
  SELECT Vendor_Nos FROM ITEMS_TABLE GROUP BY Vendor_Nos HAVING COUNT(*) > 1
  );
- Output:

	Item_Description	Vendor_Name
1	Travis Hasse Apple Pie	Mhw Ltd
2	Oak Cross Whisky	Mhw Ltd

12. Find the average Manufacturing Price for Product in each Country and only include those Country and Product combinations where the average is above 3.

- SELECT

```
    Country,
    Product,
    AVG(Manufacturing_Price) AS Avg_Manufacturing_Price
FROM PRODUCT_SALES_TABLE
GROUP BY Country, Product
HAVING AVG(Manufacturing_Price) > 3;
```

- Output:

	Country	Product	Avg_Manufacturing_Price
1	Germany	Carretera	176.500000
2	Canada	Montana	5.000000
3	Germany	Montana	5.000000

Super-Advance Questions:-

1. Find the item_description and bottle_price of items that have the same vendor_name as items with Item_Id 1.

-
```
SELECT Item_Description, Bottle_Price
FROM ITEMS_TABLE
WHERE Vendor_Name = (
    SELECT Vendor_Name
    FROM ITEMS_TABLE
    WHERE Item_Id = 1
);
```

- Output:

	Item_Description	Bottle_Price
1	Travis Hasse Apple Pie	9.77
2	Oak Cross Whisky	25.33

2. Create a stored procedure to retrieve all records from the ITEMS_TABLE table where bottle_price is greater than a given value.

-
```
CREATE PROCEDURE GetItemsByBottlePrice
    @MinPrice DECIMAL(10,2)
AS
BEGIN
    SELECT *
    FROM ITEMS_TABLE
    WHERE Bottle_Price > @MinPrice;
END;
```

- Run
  EXEC GetItemsByBottlePrice 10;
- Output:

	item_Id	item_description	vendor_nos	vendor_name	bottle_size	bottle_price
1	2	D'aristi Xtabentun	391	Anchor Distilling (preiss Imports)	750	14.12
2	4	Oak Cross Whisky	305	Mhw Ltd	750	25.33
3	8	Dupont Calvados Fine Reserve	403	Robert Kacher Selections	750	23.61

3. Create a stored procedure to insert a new record into the product_sales table.

- CREATE PROCEDURE InsertProductSales

```
    @Product_Id INT,
    @Country VARCHAR(50),
    @Product VARCHAR(100),
    @Units_Sold DECIMAL(10,2),
    @Manufacturing_Price DECIMAL(10,2),
    @Sale_Price DECIMAL(10,2),
    @Gross_Sales DECIMAL(12,2),
    @Sales DECIMAL(12,2),
    @COGS DECIMAL(12,2),
    @Profit DECIMAL(12,2),
    @Date DATE,
    @Month_Number INT,
    @Month_Name VARCHAR(20),
    @Year INT
AS
BEGIN
    INSERT INTO PRODUCT_SALES_TABLE
    VALUES(@Product_Id, @Country, @Product, @Units_Sold,
@Manufacturing_Price,
        @Sale_Price, @Gross_Sales, @Sales, @COGS, @Profit,
        @Date, @Month_Number, @Month_Name, @Year);
END;
```

4. Create a trigger to automatically update the Gross_Sales field in the product_sales table whenever Units_Sold or Sale_Price is updated.

- CREATE TRIGGER trg_UpdateGrossSales

```
ON PRODUCT_SALES_TABLE
AFTER UPDATE
AS
BEGIN
    UPDATE P
    SET Gross_Sales = P.Units_Sold * P.Sale_Price
```

```
    FROM PRODUCT_SALES_TABLE P
    INNER JOIN inserted I
        ON P.Product_Id = I.Product_Id;
END;
```

5. Write a query to find all item_description in the ITEMS_TABLE table that contain the word "Whisky" regardless of case.

- SELECT Item_Description FROM ITEMS_TABLE WHERE Item_Description LIKE '%whisky%';
- Output:

	Item_Description
1	Oak Cross Whisky

6. Write a query to find the Country and Product where the Profit is greater than the average Profit of all products.

- SELECT Country, Product
  FROM PRODUCT_SALES_TABLE
  WHERE Profit > (SELECT AVG(Profit) FROM PRODUCT_SALES_TABLE);
- Output:

	Country	Product
1	Germany	Carretera

7. Write a query to join the ITEMS_TABLE and product_sales tables on a common field (e.g., vendor_nos) and select relevant fields from both tables.

- SELECT
  I.Item_Id,
  I.Item_Description,
  I.Vendor_Nos,
  P.Product,
  P.Country,
  P.Sales
  FROM ITEMS_TABLE I
  INNER JOIN PRODUCT_SALES_TABLE P
  ON I.Vendor_Nos = P.Product_Id;
- Output:

	Item_Id	Item_Description	Vendor_Nos	Product	Country	Sales

8. Write a query to combine item_description and vendor_name into a single string for each record in the ITEMS_TABLE table, separated by a hyphen.

- SELECT
  Item_Id,
  Item_Description + ' - ' + Vendor_Name AS Full_Description
  FROM ITEMS_TABLE;
- Output:

	Item_Id	Full_Description
1	1	Travis Hasse Apple Pie - Mhw Ltd
2	2	D'aristi Xtabentun - Anchor Distilling (preiss Imports)
3	3	Hiram Walker Peach Brandy - Pernod Ricard Usa/au...
4	4	Oak Cross Whisky - Mhw Ltd
5	5	Uv Red(cherry) Vodka - Phillips Beverage Company
6	6	Heaven Hill Old Style White Label - Heaven Hill Distill...
7	7	Hyde Herbal Liqueur - Fire Tail Brands Llc
8	8	Dupont Calvados Fine Reserve - Robert Kacher Sel...

9. Write a query to display the bottle_price rounded to one decimal place for each record in the ITEMS_TABLE table.

- SELECT
  Item_Id,
  Item_Description,
  ROUND(Bottle_Price, 1) AS Bottle_Price_1_Decimal
  FROM ITEMS_TABLE;
- Output:

	Item_Id	Item_Description	Bottle_Price_1_Decimal
1	1	Travis Hasse Apple Pie	9.80
2	2	D'aristi Xtabentun	14.10
3	3	Hiram Walker Peach Brandy	6.50
4	4	Oak Cross Whisky	25.30
5	5	Uv Red(cherry) Vodka	2.00
6	6	Heaven Hill Old Style White Label	6.40
7	7	Hyde Herbal Liqueur	5.10
8	8	Dupont Calvados Fine Reserve	23.60

10. Write a query to calculate the number of days between the current date and the Date field for each record in the product_sales table.

- SELECT
  Product_Id,
  DATEDIFF(DAY, Date, GETDATE()) AS Days_Difference
  FROM PRODUCT_SALES_TABLE;
- Output:

	Product_Id	Days_Difference
1	2	4448
2	3	4297
3	6	4114
4	7	2197
5	8	1740